11. BST Inorder Traversal

Problem Statement

Write Recursive Approach for Inorder Traversal of Binary Tree.

Given a binary tree, write a recursive algorithm to perform an inorder traversal of the tree and return the elements in the order they were visited.

Examples

Example 1:

```
Input:
           5
         /   \
        3     8
       / \   / \
      2   4 7   9

Output:
Inorder Traversal: [2, 3, 4, 5, 7, 8, 9]

Explanation:
The binary tree has the following structure:
           5
         /   \
        3     8
       / \   / \
      2   4 7   9

Performing an inorder traversal visits the nodes in ascending order: 2, 3, 4, 5, 7, 8, 9.
```

Example 2:

```
Input:
           10
         /    \
        6      15
       / \    / \
      3   8  12  18
           \
```

```
             9

Output:
Inorder Traversal: [3, 6, 8, 9, 10, 12, 15, 18]

Explanation:
The binary tree has the following structure:
           10
         /    \
        6      15
       / \    / \
      3   8  12  18
            \
             9

Performing an inorder traversal visits the nodes in ascending order: 3, 6, 8, 9, 10, 12, 15,
18.
```

Example 3:

```
Input:
           20
         /    \
        12     25
       / \    / \
      8   15  22  28
     / \       \
    6   10      24

Output:
Inorder Traversal: [6, 8, 10, 12, 15, 20, 22, 24, 25, 28]

Explanation:
The binary tree has the following structure:
           20
         /    \
        12     25
       / \    / \
      8   15  22  28
     / \       \
    6   10      24

Performing an inorder traversal visits the nodes in ascending order: 6, 8, 10, 12, 15, 20,
22, 24, 25, 28.
```

Try it yourself

Try solving this question here:

Java

```
1   import java.util.*;
2
3   /*class TreeNode {
4       int val;
5       TreeNode left;
6       TreeNode right;
```

```
7
8       TreeNode(int val) {
9           this.val = val;
10      }
11  }*/
12
13  class Solution {
14      public List<Integer> inorderTraversal(TreeNode root) {
15          List<Integer> result = new ArrayList<>();
16          inorder(root, result);
17          return result;
18      }
19
20      private void inorder(TreeNode node, List<Integer> result) {
21           // TODO: Write your code here
22      }
23  }
24
```

Test Case | Run Code Result | Stdout

```
[5,3,8,2,4,7,9]
```

Use Example Testcases | Run | Submit

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Solution: Inserting A New Node In A BST

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Solution: BST Inorder Traversal

Mark as Completed